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Integration Reference Point (IRP) overview and usage guide
(3GPP TS 32.103 version 10.1.0 Release 10)**



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Foreword

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Introduction

This specification provides the IRP Overview for 3GPP specifications, addressing Interface IRPs, NRM IRPs, Measurment / Trace /SON specifications, as well as their relationships with each other.

1 Scope

The present document gives an overview about 3GPP's management interface capabilities and related functionality. It provides high-level information about IRP Framework, available IRP's as well as their relationship with each other. This document is intended to be a guide into the world of 3GPP management specifications, enabling also non-subject matter experts to to gain an understanding about 3GPP's management solutions.

In addition Annex A provides IRP packaging suggestions, intended to guide service providers as well as equipment and solutions providers to identify and choose suitable standardized management interface capabilities.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 32.150: "Telecommunication management; Integration Reference Point (IRP) Concept and definitions".

[3] 3GPP TS 32.101: "Telecommunication management; Principles and high level requirements".

[4] 3GPP TS 32.102: "Telecommunication management; Architecture".

[5] 3GPP TS 32.111-1: "Telecommunication management; Fault Management; Part 1: 3G fault management requirements".

[6] 3GPP TS 32.111-2: "Telecommunication management; Fault Management; Part 2: Alarm Integration Reference Point (IRP): Information Service (IS)".

[7] 3GPP TS 32.111-6: "Telecommunication management; Fault Management; Part 6: Alarm Integration Reference Point (IRP): Solution Set (SS) definitions".

[8] 3GPP TS 32.121: "Telecommunication management; Advanced Alarm Management (AAM) Integration Reference Point (IRP): Requirements".

[9] 3GPP TS 32.122: "Telecommunication management; Advanced Alarm Management (AAM) Integration Reference Point (IRP): Information Service (IS)".

[10] 3GPP TS 32.126: "Telecommunication management; Advanced Alarm Management (AAM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[11] 3GPP TS 32.140: "Telecommunication management; Subscription Management (SuM) requirements".

[12] 3GPP TS 32.141: "Telecommunication management; Subscription Management (SuM) architecture".

[13] 3GPP TS 32.151: "Telecommunication management; Integration Reference Point (IRP) Information Service (IS) template".

[14] 3GPP TS 32.152: "Telecommunication management; Integration Reference Point (IRP) Information Service (IS) Unified Modelling Language (UML) repertoire".

[15] 3GPP TS 32.153: "Telecommunication management; Integration Reference Point (IRP) technology specific templates, rules and guidelines".

[16] 3GPP TS 32.154: "Telecommunication management; Backward and Forward Compatibility (BFC); Concept and definitions".

[17] 3GPP TS 32.155: "Telecommunication management; Requirements template".

[18] 3GPP TS 32.171: "Telecommunication management; Subscription Management (SuM) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[19] 3GPP TS 32.172: "Telecommunication management; Subscription Management (SuM) Network Resource Model (NRM) Integration Reference Point (IRP): Information Service (IS)".

[20] 3GPP TS 32.176: "Telecommunication management; Subscription Management (SuM) Network Resource Model (NRM) Integration Reference Point (IRP); Common Object Request Broker Architecture (CORBA) Solution Set (SS) definitions".

[21] 3GPP TS 32.300: "Telecommunication management; Configuration Management (CM); Name convention for Managed Objects".

[22] 3GPP TS 32.301: "Telecommunication management; Configuration Management (CM); Notification Integration Reference Point (IRP); Requirements".

[23] 3GPP TS 32.302: "Telecommunication management; Configuration Management (CM); Notification Integration Reference Point (IRP); Information Service (IS)".

[24] 3GPP TS 32.306: "Telecommunication management; Configuration Management (CM); Notification Integration Reference Point (IRP): Solution Set (SS) definitions".

[25] 3GPP TS 32.311: "Telecommunication management; Generic Integration Reference Point (IRP) management; Requirements".

[26] 3GPP TS 32.312: "Telecommunication management; Generic Integration Reference Point (IRP) management; Information Service (IS)".

[27] 3GPP TS 32.316: "Telecommunication management; Generic Integration Reference Point (IRP) management; Solution Set (SS) definitions".

[28] 3GPP TS 32.321: "Telecommunication management; Test management Integration Reference Point (IRP); Requirements".

[29] 3GPP TS 32.322: "Telecommunication management; Test management Integration Reference Point (IRP); Information Service (IS) ".

[30] 3GPP TS 32.326: "Telecommunication management; Test management Integration Reference Point (IRP): Solution Set (SS) definitions".

[31] 3GPP TS 32.331: "Telecommunication management; Notification Log (NL) Integration Reference Point (IRP); Requirements",

[32] 3GPP TS 32.332: "Telecommunication management; Notification Log (NL) Integration Reference Point (IRP); Information Service (IS)".

[33] 3GPP TS 32.336: "Telecommunication management; Notification Log (NL) Integration Reference Point (IRP): Solution Set (SS) definitions".

[34] 3GPP TS 32.341: "Telecommunication management; File Transfer (FT) Integration Reference Point (IRP); Requirements".

[35] 3GPP TS 32.342: "Telecommunication management; File Transfer (FT) Integration Reference Point (IRP); Information Service (IS)".

[36] 3GPP TS 32.346: "Telecommunication management; File Transfer (FT) Integration ReferencePoint (IRP): Solution Set (SS) definitions".

[37] 3GPP TS 32.351: "Telecommunication management; Communication Surveillance (CS) Integration Reference Point (IRP); Requirements".

[38] 3GPP TS 32.352: "Telecommunication management; Communication Surveillance (CS) Integration Reference Point (IRP); Information Service (IS)".

[39] 3GPP TS 32.356: "Telecommunication management; Communication Surveillance (CS) Integration Reference Point (IRP); Solution Set (SS) definitions".

[40] 3GPP TS 32.361: "Telecommunication management; Entry Point (EP) Integration Reference Point (IRP); Requirements".

[41] 3GPP TS 32.362: "Telecommunication management; Entry Point (EP) Integration Reference Point (IRP); Information Service (IS)".

[42] 3GPP TS 32.366: "Telecommunication management; Entry Point (EP) Integration Reference Point (IRP); Solution Set (SS) definitions".

[43] 3GPP TS 32.381: "Telecommunication management; Partial Suspension of Itf-N Integration Reference Point (IRP); Requirements".

[44] 3GPP TS 32.382: "Telecommunication management; Partial Suspension of Itf-N Integration Reference Point (IRP); Information Service (IS)".

[45] 3GPP TS 32.386: "Telecommunication management; Partial Suspension of Itf-N Integration Reference Point (IRP); Solution Set (SS) definitions".

[46] 3GPP TS 32.391: "Telecommunication management; Delta synchronization Integration Reference Point (IRP); Requirements".

[47] 3GPP TS 32.392: "Telecommunication management; Delta synchronization Integration Reference Point (IRP); Information Service (IS)".

[48] 3GPP TS 32.396: "Telecommunication management; Delta synchronization Integration Reference Point (IRP); Solution Set (SS) definitions".

[49] 3GPP TS 32.401: "Telecommunication management; Performance Management (PM); Concept and requirements".

[50] 3GPP TS 32.405: "Telecommunication management; Performance Management (PM); Performance measurements; Universal Terrestrial Radio Access Network (UTRAN)".

[51] 3GPP TS 32.406: "Telecommunication management; Performance Management (PM); Performance measurements; Core Network (CN) Packet Switched (PS) domain".

[52] 3GPP TS 32.407: "Telecommunication management; Performance Management (PM); Performance measurements; Core Network (CN) Circuit Switched (CS) domain; UMTS and combined UMTS/GSM".

[53] 3GPP TS 32.408: "Telecommunication management; Performance Management (PM); Performance measurements; Teleservice".

[54] 3GPP TS 32.409: "Telecommunication management; Performance Management (PM); Performance measurements; IP Multimedia Subsystem (IMS)".

[55] 3GPP TS 32.410: "Telecommunication management; Key Performance Indicators (KPI) for UMTS and GSM".

[56] 3GPP TS 32.411: "Telecommunication management; Performance Management (PM) Integration Reference Point (IRP): Requirements".

[57] 3GPP TS 32.412: "Telecommunication management; Performance Management (PM) Integration Reference Point (IRP): Information Service (IS)".

[58] 3GPP TS 32.416: "Telecommunication management; Performance Management (PM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[59] 3GPP TS 32.421: "Telecommunication management; Subscriber and equipment trace; Trace concepts and requirements".

[60] 3GPP TS 32.422: "Telecommunication management; Subscriber and equipment trace; Trace control and configuration management".

[61] 3GPP TS 32.423: "Telecommunication management; Subscriber and equipment trace; Trace data definition and management

[62] 3GPP TS 32.425: "Telecommunication management; Performance Management (PM); Performance measurements Evolved Universal Terrestrial Radio Access Network (E-UTRAN)".

[63] 3GPP TS 32.426: "Telecommunication management; Performance Management (PM); Performance measurements Evolved Packet Core (EPC) network".

[64] 3GPP TS 32.432: "Telecommunication management; Performance measurement: File format definition".

[65] 3GPP TS 32.435: "Telecommunication management; Performance measurement; eXtensible Markup Language (XML) file format definition".

[66] 3GPP TS 32.436: "Telecommunication management; Performance measurement: Abstract Syntax Notation 1 (ASN.1) file format definition".

[67] 3GPP TS 32.441: "Telecommunication management; Trace Management Integration Reference Point (IRP); Requirements".

[68] 3GPP TS 32.442: "Telecommunication management; Trace Management Integration Reference Point (IRP); Information Service (IS)".

[69] 3GPP TS 32.446: "Telecommunication management; Trace Management Integration Reference Point (IRP); Solution Set (SS) definitions".

[70] 3GPP TS 32.451: "Telecommunication management; Key Performance Indicators (KPI) for Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Requirements".

[71] 3GPP TS 32.452: "Telecommunication management; Performance Management (PM); Performance measurements Home Node B (HNB) Subsystem (HNS)".

[72] 3GPP TS 32.453: "Telecommunication management; Performance Management (PM); Performance measurements Home enhanced Node B (HeNB) Subsystem (HeNS)".

[73] 3GPP TS 32.454: "Telecommunication management; Key Performance Indicators (KPI) for the IP Multimedia Subsystem (IMS); Definitions".

[74] 3GPP TS 32.455: "Telecommunication management; Key Performance Indicators (KPI) for the Evolved Packet Core (EPC); Definitions".

[75] 3GPP TS 32.500: "Telecommunication management; Self-Organizing Networks (SON); Concepts and requirements".

[76] 3GPP TS 32.501: "Telecommunication management; Self-configuration of network elements; Concepts and requirements".

[77] 3GPP TS 32.502: "Telecommunication management; Self-configuration of network elements Integration Reference Point (IRP); Information Service (IS)".

[78] 3GPP TS 32.506: "Telecommunication management; Self-configuration of network elements Integration Reference Point (IRP); Solution Set (SS) definitions".

[79] 3GPP TS 32.511: "Telecommunication management; Automatic Neighbour Relation (ANR) management; Concepts and requirements".

[80] 3GPP TS 32.521: "Telecommunication management; Self-Organizing Networks (SON) Policy Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[81] 3GPP TS 32.522: "Telecommunication management; Self-Organizing Networks (SON) Policy Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[82] 3GPP TS 32.526: "Telecommunication management; Self-Organizing Networks (SON); Policy Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[83] 3GPP TS 32.531: "Telecommunication management; Software management (SwM); Concepts and Integration Reference Point (IRP) Requirements".

[84] 3GPP TS 32.532: "Telecommunication management; Software management (SwM); Integration Reference Point (IRP); Information Service (IS)".

[85] 3GPP TS 32.536: "Telecommunication management; Software management Integration Reference Point (IRP); Solution Set (SS) definitions".

[86] 3GPP TS 32.541: "Telecommunication management; Self-Organizing Networks (SON); Self-healing concepts and requirements".

[87] 3GPP TS 32.551: "Telecommunication management; Energy Saving Management (ESM); Concepts and requirements".

[88] 3GPP TS 32.600: "Telecommunication management; Configuration Management (CM); Concept and high-level requirements".

[89] 3GPP TS 32.601: "Telecommunication management; Configuration Management (CM); Basic CM Integration Reference Point (IRP); Requirements".

[90] 3GPP TS 32.602: "Telecommunication management; Configuration Management (CM); Basic CM Integration Reference Point (IRP); Information Service (IS)".

[91] 3GPP TS 32.606: "Telecommunication management; Configuration Management (CM); Basic CM Integration Reference Point (IRP); Solution Set (SS) definitions".

[92] 3GPP TS 32.611: "Telecommunication management; Configuration Management (CM); Bulk CM Integration Reference Point (IRP): Requirements".

[93] 3GPP TS 32.612: "Telecommunication management; Configuration Management (CM); Bulk CM Integration Reference Point (IRP): Information Service (IS)".

[94] 3GPP TS 32.616: "Telecommunication management; Configuration Management (CM); Bulk CM Integration Reference Point (IRP); Solution Set (SS) definitions".

[95] 3GPP TS 32.621: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[96] 3GPP TS 32.622: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[97] 3GPP TS 32.626: "Telecommunication management; State management data definition Integration Reference Point (IRP); Solution Set (SS) definitions".

[98] 3GPP TS 32.631: "Telecommunication management; Configuration Management (CM); Core network resources Integration Reference Point (IRP); Requirements".

[99] 3GPP TS 32.632: "Telecommunication management; Configuration Management (CM); Core Network Resources Integration Reference Point (IRP); Network Resource Model (NRM)".

[100] 3GPP TS 32.636: "Telecommunication management; Configuration Management (CM); Core network resources Integration Reference Point (IRP); Solution Set (SS) Definitions".

[101] 3GPP TS 32.641: "Telecommunication management; Configuration Management (CM); UTRAN network resources Integration Reference Point (IRP); Requirements".

[102] 3GPP TS 32.642: "Telecommunication management; Configuration Management (CM); UTRAN network resources Integration Reference Point (IRP); Network Resource Model (NRM)".

[103] 3GPP TS 32.646: "Telecommunication management; Configuration Management (CM); UTRAN network resources Integration Reference Point (IRP); Solution Set (SS) definitions".

[104] 3GPP TS 32.651: "Telecommunication management; Configuration Management (CM); GERAN network resources Integration Reference Point (IRP); Requirements".

[105] 3GPP TS 32.652: "Telecommunication management; Configuration Management (CM); GERAN network resources Integration Reference Point (IRP); Network Resource Model (NRM)".

[106] 3GPP TS 32.656: "Telecommunication management; Configuration Management (CM); GERAN network resources Integration Reference Point (IRP); Solution Set (SS) definitions".

[107] 3GPP TS 32.661: "Telecommunication management; Configuration Management (CM); Kernel CM Requirements".

[108] 3GPP TS 32.662: "Telecommunication management; Configuration Management (CM); Kernel CM Information Service (IS)".

[109] 3GPP TS 32.671: "Telecommunication management; Configuration Management (CM); State Management Integration Reference Point (IRP); Requirements".

[110] 3GPP TS 32.672: "Telecommunication management; Configuration Management (CM); State Management Integration Reference Point (IRP); Information Service (IS)".

[111] 3GPP TS 32.673: "Telecommunication management; Configuration Management (CM); State Management Integration Reference Point (IRP); Common Object Request Broker Architecture (CORBA) Solution Set (SS)".

[112] 3GPP TS 32.690: "Telecommunication management; Inventory Management (IM); Requirements".

[113] 3GPP TS 32.691: "Telecommunication management; Inventory Management (IM) network resources Integration Reference Point (IRP): Requirements".

[114] 3GPP TS 32.692: "Telecommunication management; Inventory Management (IM) network resources Integration Reference Point (IRP); Network Resource Model (NRM)".

[115] 3GPP TS 32.696: "Telecommunication management; Inventory Management (IM) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[116] 3GPP TS 32.711: "Telecommunication management; Configuration Management (CM); Transport Network (TN) interface Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[117] 3GPP TS 32.712: "Telecommunication management; Configuration Management (CM); Transport Network (TN) interface Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[118] 3GPP TS 32.716: "Telecommunication management; Configuration Management (CM); Transport Network (TN) interface Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[119] 3GPP TS 32.721: "Telecommunication management; Configuration Management (CM); Repeater network resources Integration Reference Point (IRP); Requirements".

[120] 3GPP TS 32.722: "Telecommunication management; Configuration Management (CM); Repeater network resources Integration Reference Point (IRP); information Service (IS)".

[121] 3GPP TS 32.726: "Telecommunication management; Configuration Management (CM); Repeater network resources Integration Reference Point (IRP); Solution Set (SS) definitions".

[122] 3GPP TS 32.731: "Telecommunication management; IP Multimedia Subsystem (IMS) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[123] 3GPP TS 32.732: "Telecommunication management; IP Multimedia Subsystem (IMS) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[124] 3GPP TS 32.736: "Telecommunication management; IP Multimedia Subsystem (IMS) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[125] 3GPP TS 32.741: "Telecommunication management; Configuration Management (CM); Signalling Transport Network (STN) interface Network Resource Model (NRM) IntegrationReference Point (IRP); Requirements".

[126] 3GPP TS 32.742: "Telecommunication management; Configuration Management (CM); Signalling Transport Network (STN) interface Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[127] 3GPP TS 32.746: "Telecommunication management; Configuration Management (CM); Signalling Transport Network (STN) interface Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[128] 3GPP TS 32.751: "Telecommunication management; Evolved Packet Core (EPC) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[129] 3GPP TS 32.752: "Telecommunication management; Evolved Packet Core (EPC) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[130] 3GPP TS 32.756: "Telecommunication management; Evolved Packet Core (EPC) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) Definitions".

[131] 3GPP TS 32.761: "Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[132] 3GPP TS 32.762: "Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[133] 3GPP TS 32.766: "Telecommunication management; Evolved Universal Terrestrial RadioAccess Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[134] 3GPP TS 32.771: "Telecommunication management; Home Node B (HNB) Subsystem (HNS); Network Resource Model (NRM); Integration Reference Point (IRP); Requirements".

[135] 3GPP TS 32.772: "Telecommunication management; Home Node B (HNB) Subsystem (HNS); Network Resource Model (NRM); Integration Reference Point (IRP); Information Service (IS)".

[136] 3GPP TS 32.776: "Telecommunication management; Home Node B (HNB) Subsystem (HNS); Network Resource Model (NRM); Integration Reference Point (IRP); Solution Set (SS) definitions".

[137] 3GPP TS 32.781: "Telecommunication management; Home enhanced Node B (HeNB) Subsystem (HeNS); Network Resource Model (NRM); Integration Reference Point (IRP); Requirements".

[138] 3GPP TS 32.782: "Telecommunication management; Home enhanced Node B (HeNB) Subsystem (HeNS); Network Resource Model (NRM); Integration Reference Point (IRP): Information Service (IS)".

[139] 3GPP TS 32.786: "Telecommunication management; Home enhanced Node B (HeNB) Subsystem (HeNS); Network Resource Model (NRM); Integration Reference Point (IRP); Solution Set (SS) definitions".

[140] 3GPP TS 52.402: "Telecommunication management; Performance Management (PM); Performance measurements – GSM".

[141] 3GPP TS 32.181: "Telecommunication management; User Data Convergence (UDC); Framework for Model Handling and Management".

[142] 3GPP TS 32.182: "Telecommunication management; User Data Convergence (UDC); Common baseline information model (CBIM)".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1], TS 32.150 [2].

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] apply.

4 IRP Framework

4.1 Introduction

For the purpose of management interface development 3GPP has developed an interface concept known as Integration Reference Point (IRP) to promote the wider adoption of standardized management interfaces in telecommunication networks. The IRP concept and associated methodology employs model-driven development, protocol and technology neutral modelling methods as well as protocol specific solution sets to achieve its goals.

IRP Framework and Methodology related 3GPP specifications:

- 32.150 [2]: Integration Reference Point (IRP) Concept and definitions
- 32.151 [13]: IRP Information Service (IS) template
- 32.152 [14]: IRP IS Unified Modeling Language (UML) repertoire
- 32.153 [15]: IRP technology specific templates, rules and guidelines
- 32.154 [16]: Backward and Forward Compatibility Concept and definitions
- 32.155[17]: Requirements template

4.2 IRP Framework Highlights

4.2.1 IRP Concept

IRP specifications are specified using a 3-level approach: Requirements, IS-level and SS-level.

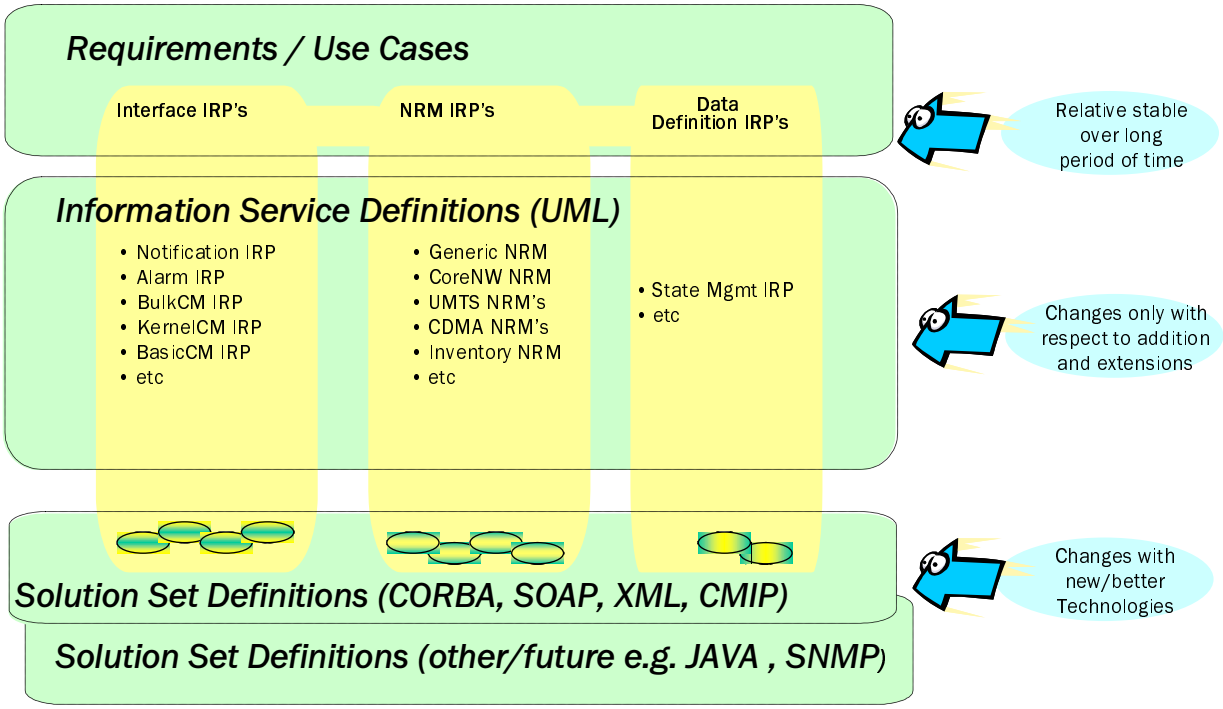


Figure 4.2-1: The IRP 3-Level Specifications Approach combined with the three IRP categories [2].

Requirements-level:

The "Requirements-level" intends to provide conceptual and use cases definitions for a specific management interface aspect as well as defining subsequent requirements for this IRP.

IS-level:

- The "IS-level" provides the technology independent specification of an IRP.

SS-level:

The "SS-level" finally provides the mapping of IS definitions into one or more technology-specific Solution Sets. This concept provides support for multiple interface technologies as applicable on a vendor and/or network type basis and also enables accommodation of future interface technologies - without the need to redefine requirements and IS-level definitions.

4.2.2 Interface IRP vs. NRM IRP

There are three categories of IRP specifications:

- Interface IRPs
- NRM IRPs
- Data Definition IRPs

Each category is partitioned into Requirements, IS-level and SS-level specifications.

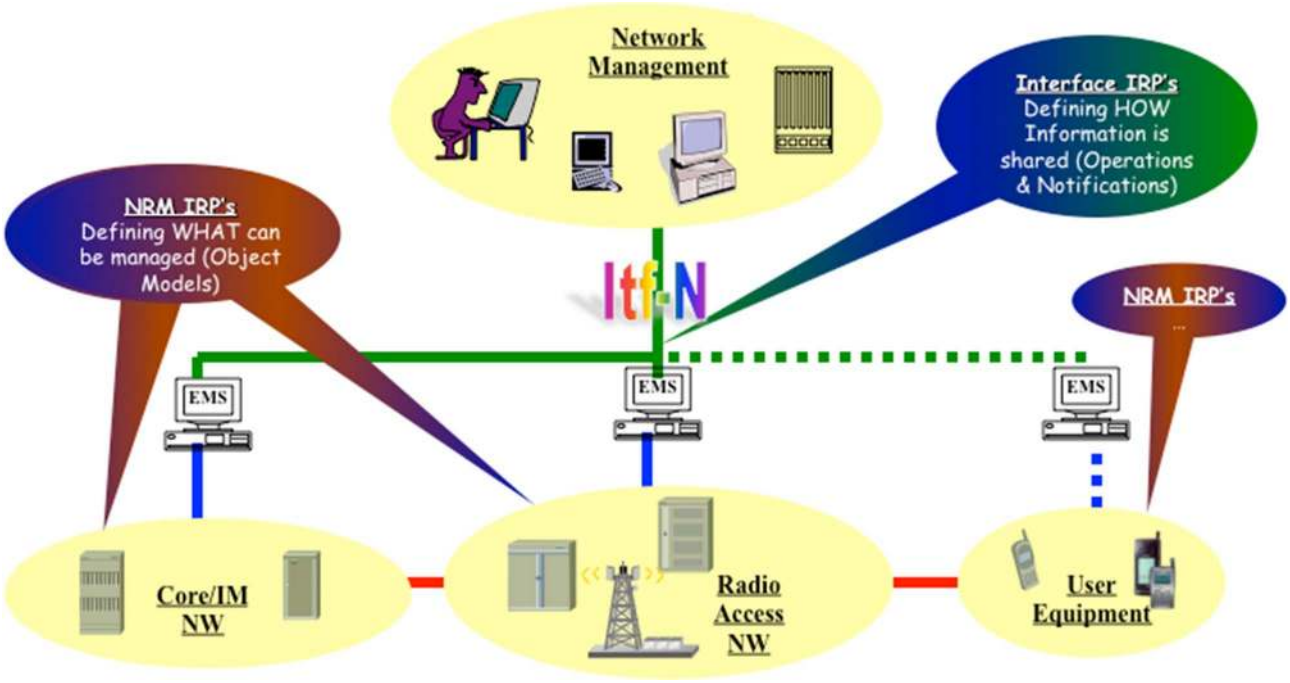


Figure 4.2-2: Relationship Interface IRP vs NRM IRP

4.2.3 IRP Development Principles

NRM IRP Extendibility - Enabling Technology, Organizational & Vendor-specific NRM extensions via

- Rule-based NRM Extensions (e.g. Sub-classing)
- vsDataContainer

Interface IRP Flexibility - Enabling: wide applicability, phased introduction capabilities & broad industry adoption (not just for wireless) via

- Flexible use of qualifiers "mandatory", "optional", "conditional" for operation, notifications and/or parameters
- NRM/Technology-neutrality & avoiding competing procedures

4.2.4 IRP Specification Structure

All IRP's follow a uniform specification structure:

- xx1: xyz IRP Requirements
- xx2: xyz IRP IS
- *xx3: xyz IRP CORBA SS (merged into xx6 after R9)*
- *xx4: xyz IRP CMIP SS (retired after R6)*
- *xx5: xyz IRP XML Definitions (applicable for NRM's & notification emitting IRP's - merged into xx6 after R9)*
- xx6: xyz IRP SS (since R10, containing all technology specific Solution Set definitions)
- *xx7: xyz IRP SOAP SS (merged into xx6 after R9)*

4.2.5 Essential IRP Definitions [2]

Data Definition IRP: 3GPP publishes IRP specifications relating to commonly used data definitions that can be imported for use by Interface IRP and/or NRM IRP. This term represents all such specifications. An example of a Data Definition IRP is the State Management IRP (32.671 [109], 32.672 [110], 32.673 [111], etc).

Integration Reference Point (IRP): An architectural concept that is described by a set of specifications for definition of a certain aspect of a management interface, comprising a **Requirements** specification, an **Information Service** specification, and one or more **Solution Set** specifications.

Interface IRP: 3GPP publishes a number of IRP specifications each of which is related to a set of operations and notifications for a specific telecom management domain such as alarm management, configuration management, etc. Interface IRPs also contain definitions of Support IOCs. This term represents all such specifications. An example of an Interface IRP is the Basic CM IRP (the set of TSs 32.601 [894], 32.602 [90], 32.606 [91], etc.).

IRPAgent: Encapsulates a well-defined subset of network (element) functions. It interacts with IRPManagers using one or more IRPs. From the IRPManager's perspective, the IRPAgent behaviour is only visible via the IRP(s).

Information Service (IS): an IRP Information Service describes the information related to the entities (either network resources or support objects) to be managed and the way that the information may be managed for a certain functional area (e.g. the Alarm IRP Information Service in the fault management area). Information Services are defined for all IRPs.

IRPManager: Models a user of IRPAgent(s) and it interacts directly with the IRPAgent(s) using IRP(s).
Since the IRPManager represents an IRPAgent user, it gives a clear picture of what the IRPAgent is supposed to do. From the IRPAgent perspective, the IRPManager behaviour is only visible via the IRP.

Network Resource Model (NRM): An Information Service describing Information Object Classes representing the manageable aspects of network resources, e.g. an RNC or NodeB.

NRM IRP: 3GPP publishes a number of IRP specifications each of which is related to a particular NRM (Network Resource Model) as defined in 3GPP TS 32.101 [3]. NRM IRPs do not define any operations or notifications. This term represents all such specifications. Note: In some NRM IRP titles, for historic reasons, they are named "…network resources IRP"). An example of an NRM IRP is the Generic NRM IRP (32.621 [95], 32.622 [96], 32.623 [97], etc.).

Solution Set (SS): contains a mapping of the IRP Information Service (IS) defined entities (that are technology-agnostic) to technology specific termed entities. It does not contain specification of the entities' semantics. The semantics can be found in the corresponding IS. It is noted that one IS can be mapped to one or several SSs.

4.3 IRP related High-level Requirements & Architecture Specifications

The IRP Framework has been developed in response to management requirements documented in the following specifications:

- 32.101 [3]: Principles and high level requirements
- 32.102[4]: Architecture
- 32.111-1 [5]: 3G fault management requirements
- 32.140 [11], 32.141 [12]: SuM requirements / SuM architecture
- 32.300 [21]: Name convention for Managed Objects
- 32.401[49]: PM Concept and requirements
- 32.500 [75]: SON Concepts and requirements
- 32.511 [79]: ANR management Concepts and requirements
- 32.521 [80]: Self-optimization Concepts and requirements
- 32.541 [86]: Self-healing Concepts and requirements
- 32.551 [87]: Energy Saving Management (ESM) Concepts and requirements
- 32.600 [88]: CM Concept and high-level requirements

5 Interface IRP's

5.1 Interface IRP Overview

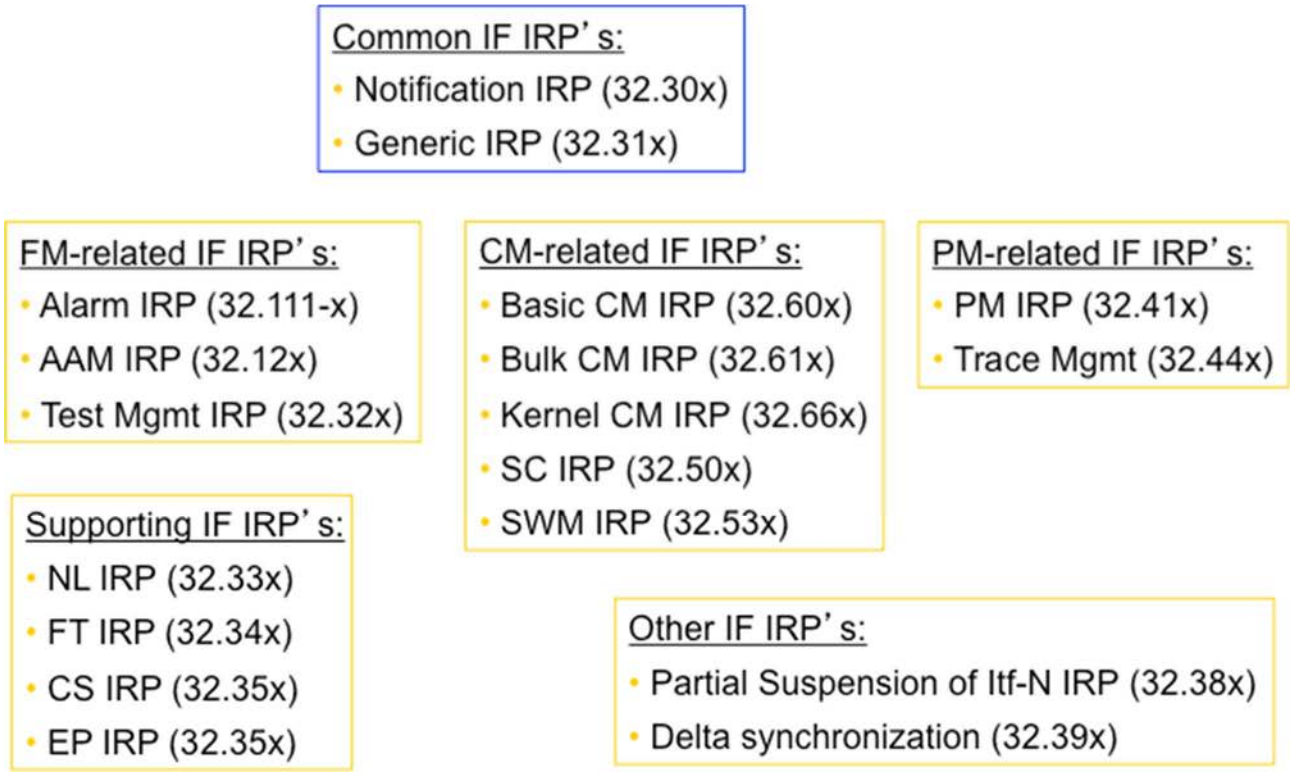


Figure 5.1-1: Interface IRP's 3GPP R10

5.2 Generic/Common Interface IRP's

Notification IRP – 32.300 [21]**, 32.301**[22]**, 32.302**[23]**, 32.306**[24]

The purpose of the Notification IRP is to define an interface through which an IRPManager can subscribe to an IRPAgent for receiving notifications, manage these subscriptions, and optionally set filters.

Generic IRP – 32.311 [25]**, 32.312** [26 **32.316** [27]

This IRP defines common services supported by all IRPs such as AlarmIRP. With this common service supported by all IRPs, an IRPManager can retrieve the profile of operations and notifications supported by a given IRP name-contained by an IRPAgent. An IRPManager can also retrieve the IRPVersions supported by a given IRP.

5.3 FM related Interface IRP's

Alarm IRP – 32.111-1 [5], 32.111-2 [6], 32.111-6[7]

The Alarm (IRP) addresses the alarm surveillance aspects of Fault Management (FM), applied to Itf-N. It defines alarm semantics & syntax as well as alarm management related operations and notifications.

Advanced Alarm Management IRP – 32.121 [8]**, 32.122** [9]**, 32.126** [10]

The purpose of Advanced Alarm Management (AAM) IRP is to define an interface through which an IRPManager can categorize alarm notifications. It defines, for the purpose of categorizing alarm notifications, the information observable and controlled by management system's client and it also specifies the semantics of the interactions used to carry this information.

Test management IRP – 32.321 [28]**, 32.322** [29]**, 32.326**[30]

The purpose of the Test management IRP is to define an interface through which an IRPManager can manage and monitor tests in NE's, as well as methods to receive test results.

5.4 CM related Interface IRP's

Self-Configuration IRP – 32.500 [75]**, 32.501** [76]**, 32.502** [77]**, 32.506** [78]

The Self-Configuration provides methods to allow automatic configuration of eNBs, which are brought newly into a network.

Software Management IRP – 32.531 [83]**, 32.532** [84]**, 32.536** [85]

The Software Management IRP provides an IRPManager with capabilities to automate software management via Itf-N.

Basic CM IRP – 32.600 [88]**, 32.601**[89]**, 32.602** [90]**, 32.606** [91

Using this IRP, an IRPAgent can communicate basic Configuration Management related information to one or several IRPManagers. The function of this Basic CM IRP Information Service is to define an interface for the retrieval and modification of Configuration Management Information.

Bulk CM IRP – 32.611 [92]**, 32.612** [93]**, 32.616** [94]

Using the Bulk CM IRP an IRPAgent can communicate CM related information to one or several IRPManagers using bulk data transfer mehods.

- BulkCM SimpleUpload, enabling upload of resource information by the IRPManager without explicit session control
- BulkCM Controlled Upload, enabling a session controlled upload of resource information by the IRPManager
- BulkCM Controlled Upload & Provisioning, enabling a session controlled upload and provisioning of resource information by the IRPManager

Kernel CM IRP – 32.661 [107]**, 32.662** [108]

The function of this Kernel CM IRP Information Service is to define an interface that provides essential CM notification services. While it is not expected that the Kernel CM IRP alone will provide adequate CM capabilities, the Kernel CM IRP is expected to provide the common supporting CM notifications required for other CM IRPs such as the Basic CM IRP or the Bulk CM IRP, each of which require the Kernel CM IRP.

5.5 PM & Trace related Interface IRP's

Performance Management IRP – 32.410 [55]**, 32.411**[56]**, 32.412** [57]**, 32.416** [58]

The Performance Management IRP is defines an interface through which an IRPManager can manage measurement jobs as well threshold values.

Trace Management IRP – 32.441 [67]**, 32.442** [68]**, 32.446** [69]

The Trace Management IRP is defines an interface through which an IRPManager can activate and manage equipment and subscriber trace jobs.

5.6 Special-purpose related Interface IRP's

Notification Log IRP – 32.331[31]**, 32.332**[32]**, 32.336**[33]

The purpose of the Notification Log IRP is to define an interface through which an IRPManager can manage a Notification Log, as well as methods for retrieval of logged notifications.

File Transfer IRP – 32.341[34]**, 32.342**[35]**, 32.346**[36]

The purpose of the File Transfer IRP is to define a generic tools set supporting the management of file transfers between IRPManager and IRPAgent. Curently the following management data types are supported by this IRP:

- performance data files
- configuration & inventory files
- test result files
- call trace files
- notification log files
- charging files

Communication Surveillance IRP – 32.351 [37]**, 32.352** [38]**, 32.356**[39]

The purpose of the Communication Surveillance IRP is to provide an IRPManager with the capability to monitor the health of the communications link towards an IRPAgent.

Entry Point IRP – 32.361 [40]**, 32.362**[41]**, 32.366** [42]

The purpose of the Entry Point IRP is to provide an IRPManager with discovery and entry point capabilities towards an IRPAgent and its supported IRP's.

Partial Suspension of Itf-N IRP – 32.381 [43]**, 32.382** [44]**, 32.386** [45]

The purpose of Partial Suspension of Itf-N IRP is to define an interface through which an IRPManager can suspend the forwarding of notifications via Itf-N which were generated in parts of the managed systems.

Delta synchronization IRP – 32.391 [46]**, 32.392** [47]**, 32.396** [48]

The purpose of Delta Synchronization IRP is to define an interface through which an IRPManager can request only those data that have changed (i.e. changed, were created or deleted) from a synchronization point onwards.

6 NRM IRP's

6.1 NRM IRP Overview

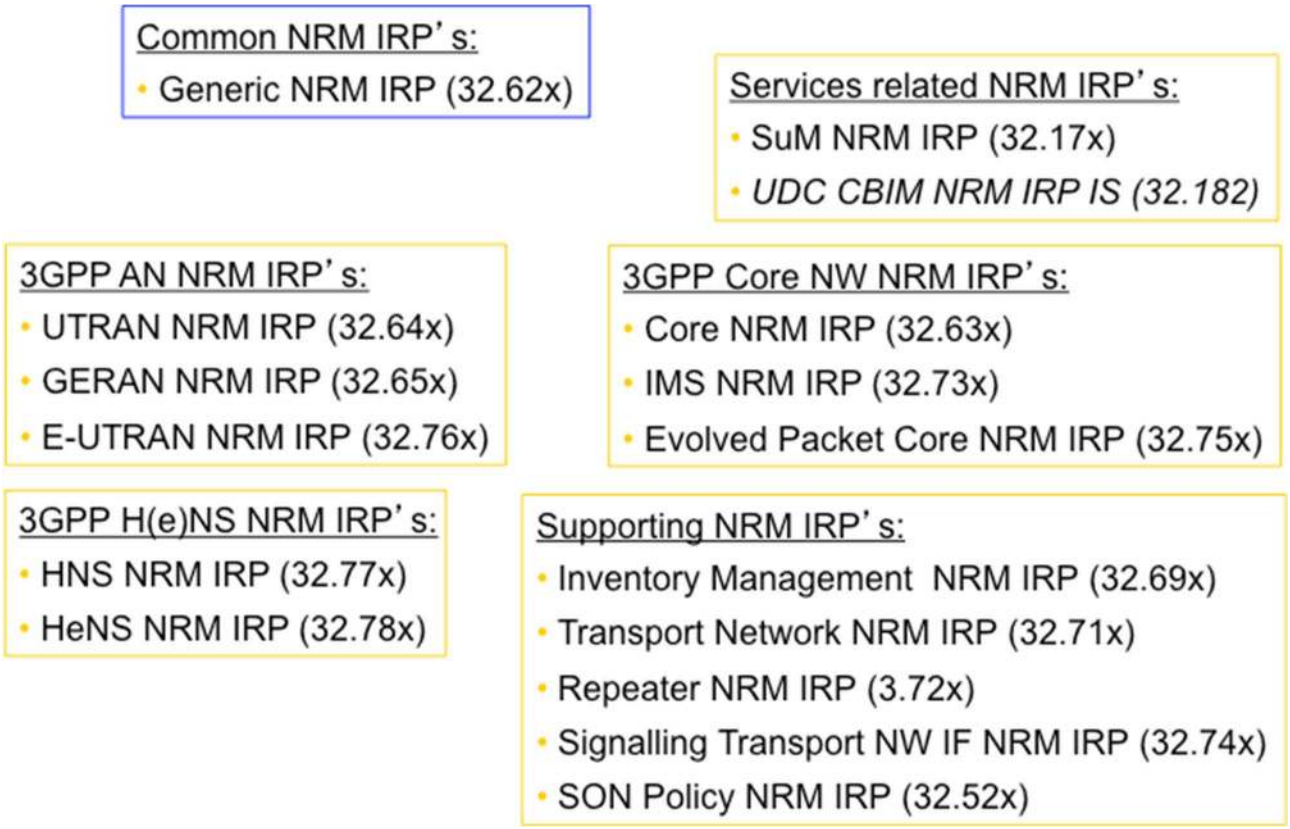


Figure 6.1-1: NRM IRP's 3GPP R10

6.2 Common NRM IRP's

Generic NRM IRP – 32.621 [95]**, 32.622** [96]**, 32.626** [97]

The Generic NRM IRP provides the generic network resources information that can be communicated between an IRPAgent and one or several IRPManagers for network management purposes.

6.3 Access Network (AN) related NRM IRP's

UTRAN NRM IRP – 32.641[101] **, 32.642** [102]**, 32.646** [103]

The UTRAN NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning UTRAN specific network resource, by reusing relevant parts of the Generic NRM IRP in 32.621 [95], 32.622 [96], 32.626 [97] series.

GERAN NRM IRP – 32.651 [104]**, 32.652** [105]**, 32.656** [106]

The GERAN NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning GERAN specific network resource, by reusing relevant parts of the Generic NRM IRP in 32.621 [95], 32.622 [96], 32.626 [97] series.

EUTRAN NRM IRP – 32.761 [131]**, 32.762** [132]**, 32.766** [133]

The EUTRAN NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning EUTRAN specific network resource, by reusing relevant parts of the Generic NRM IRP in 32.621[95], 32.622 [96], 32.626 [97] series.

6.4 Core/IMS related NRM IRP's

Core NRM IRP – 32.631 [98]**, 32.632** [99]**, 32.636** [100]

The Core NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning core network specific resource, by reusing relevant parts of the Generic NRM IRP in 32.621 [95], 32.622 [96], 32.626 [97] series.

IMS NRM IRP – 32.731 [122]**, 32.732** [123]**, 32.736** [124]

The IMS NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning IP MultiMedia Subsystem specific network resource, by reusing relevant parts of the Generic NRM IRP in 32.621 [95], 32.622 [96], 32.626 [97] series.

EPC NRM IRP – 32.751 [128]**, 32.752** [129]**, 32.756** [130]

The EPC NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Evolved Packet Core specific network resource, by reusing relevant parts of the Generic NRM IRP in 32.621 [95], 32.622 [96], 32.626 [97] series.

6.5 H(e)NB related NRM IRP's

HNS NRM IRP – 32.771 [134]**, 32.772** [135]**, 32.776** [136]

The HNS NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Home NodeB Subsystem specific network resource, by reusing relevant parts of the Generic NRM IRP in 32.621 [95], 32.622 [96], 32.626 [97] series.

HeNS NRM IRP – 32.781 [137]**, 32.782** [138]**, 32.786** [139]

The HeNS NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Home enhanced NodeB Subsystem specific network resource, by reusing relevant parts of the Generic NRM IRP in 32.621 [95], 32.622 [96], 32.626 [97] series.

6.6 Service Management related NRM IRP's and UDC specifications

SuM NRM IRP – 32.171 [18], 32.172 [19], 32.176 [20]

Subscription Management is a feature that permits Service Providers, Value Added Service Providers and Mobile Operators to provision services for a specific subscriber. The SuM NRM IRP defines the Subscription Management specific network resource information between an IRPAgent and one or several IRPManagers.

UDC – 32.181 [141], 32.182 [142]

User Data Convergence (UDC) presents a layered architecture where user data is accessible at a logically unique repository called User Data Repository (UDR). The UDC CBIM defines a Common Baseline Information Model for UDC, which denotes an abstract, formal representation of entity types that are common to many applications using the UDR.

6.7 Special-purpose related NRM IRPs

Inventory Management NRM IRP – 32.690 [112]**, 32.691**[113]**, 32.692** [114]**, 32.696** [115]

Inventory Management (IM) provides the operator with the ability to assure correct and effective operation of the telecom network as it evolves. IM actions have the objective to monitor the actual configuration on the Network Elements (NEs) and Network Resources (NRs). The Inventory Management NRM IRP defines the inventory specific network resource information between an IRPAgent and one or several IRPManagers.

SON Policy NRM IRP – 32.521 [80]**, 32.522** [81]**, 32.526** [82]

The SON Policy NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Self-Organizaing Networks Policies. Currently the following SON use cases are supported by this NRM IRP:

- SON Self-Optimization Management (requirements determined by TS 32.521[80])
- SON Self-Healing Management (requirements determined by TS 32.541[86])
- Energy Saving Management (requirements determined by TS 32.551[87])

Transport Network NRM IRP – 32.711 [116]**, 32.712** [117]**, 32.716** [118]

The Transport Network NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Transport resources, by reusing relevant parts of the Generic NRM IRP in 32.621 [95], 32.622 [96], 32.626 [97] series.

Repeater NRM IRP – 32.721 [119]**, 32.722** [120]**, 32.726** [121]

The Repeater NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Repeater resources, by reusing relevant parts of the Generic NRM IRP in 32.621 [95], 32.622 [96], 32.626 [97] series.

Signalling Transport NW IF NRM IRP – 32.741 [125]**, 32.742** [126]**, 32.746** [127]

The Signalling Transport NW IF NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Signalling Transport resources, by reusing relevant parts of the Generic NRM IRP in 32.621[95], 32.622 [96], 32.626 [97] series.

7 Measurement & Trace Definitions

7.1 Performance Measurements

7.1.1 Overview and relationships to IRP Framework

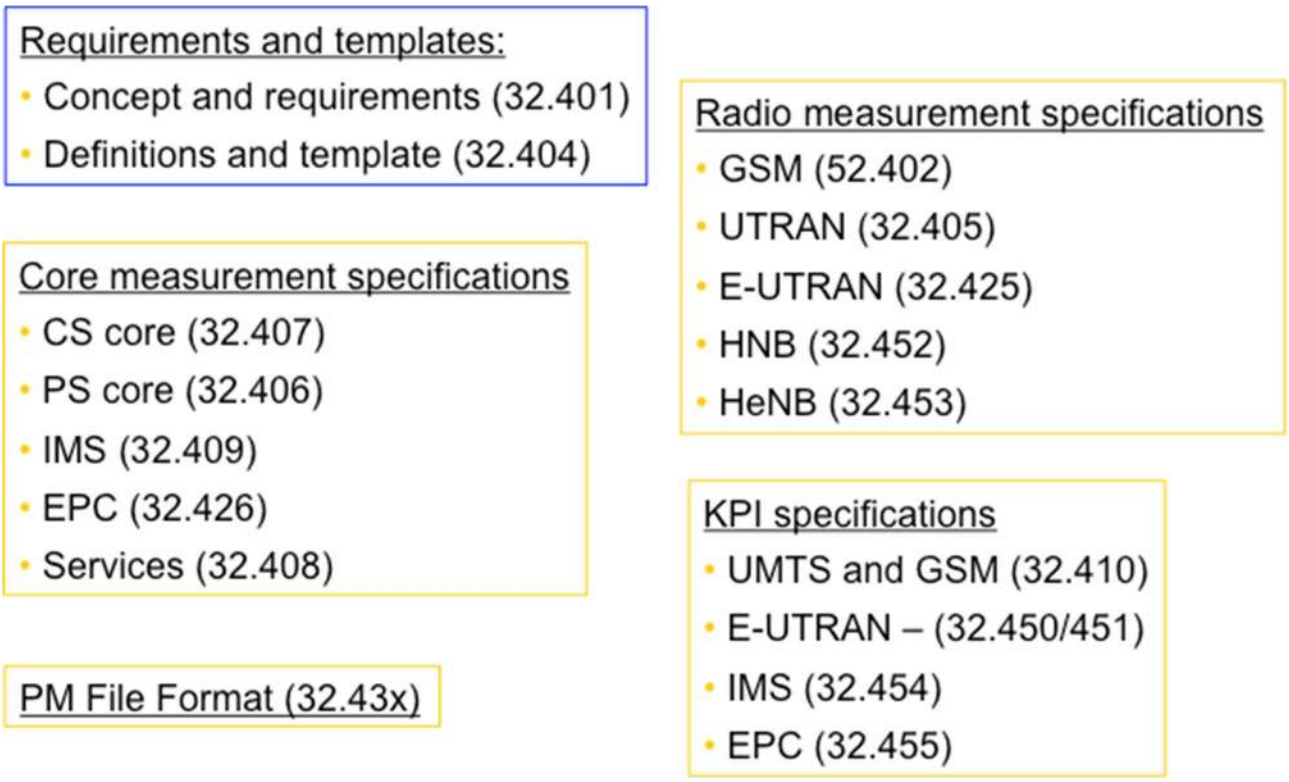


Figure 7.1.1-1: PM-related Specifications 3GPP R10

7.1.2 Performance Measurement Specifications

PS core Performance Measurements – 32.406 [51]

The purpose of this specification is to define performance measurements specific to Core Network Packet Switched Domain in a UMTS network or combined UMTS/GSM network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

CS core Performance Measurements – 32.407 [52]

The purpose of this specification is to define performance measurements specific to Core Network Circuit Switched Domain in a UMTS network or combined UMTS/GSM network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

Services Performance Measurements – 32.408 [53]

The purpose of this specification is to define performance measurements specific to teleservices(categorized by Multimedia Messageing Service, Short Message Service, Multimedia Calls, Voice Call and Intelligent Services) in a UMTS network or combined UMTS/GSM network, . The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

IMS Performance Measurements – 32.409 [54]

The purpose of this specification is to define performance measurements specific to an IMS (IP Multimdedia Subsystem) network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

EPC Performance Measurements – 32.426 [63]

The purpose of this specification is to define performance measurements specific to an EPC network or combined EPC/UMTS/GSM network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

GSM Performance Measurements – 52.402 [140]

The purpose of this specification is to define performance measurements specific to a GSM system. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.**UTRAN Performance Measurements – 32.405 [50]**

The purpose of this specification is to define performance measurements specific to a UTRAN in UMTS network or combined UMTS/GSM network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

E-UTRAN Performance Measurements – 32.425 [62]

The purpose of this specification is to define performance measurements specific to an E-UTRAN network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.**HNB Performance Measurements – 32.452 [71]**

The purpose of this specification is to define performance measurements specific to Home NodeB Subsystem(consists of HNB and HNB-GW). The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

HeNB Performance Measurements – 32.453 [72]

The purpose of this specification is to define performance measurements specific to Home enhanced NodeB Subsystem(consists of HeNB and optionally HeNB-GW). The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

UMTS & GSM KPI's – 32.410 [55]

The purpose of this specification is to define Key Performance Indicators (KPIs) for GSM and UMTS. KPI definitions include high level KPIs that are a) common across GSM and UMTS networks; and b) **specific** to network techniques such as GSM and UMTS networks.

E-UTRAN KPI's – 32.451 [70]

This specification defines requirements (business level requirements, specification level requirements and use case descriptions) related to KPIs for E-UTRAN.

IMS KPI's – 32.454 [73]

This specification defines KPIs for the IP Multimedia Subsystem (IMS).

EPC KPI's – 32.455 [74]

This specification defines KPIs for the Evolved Packet Core(EPC).network.

PM File Format – 32.432 [64]**, 32.435** [65]**, 32.436** [66]

This set of specifications describe the general semantics of performance measurement result and collection. It defines the report file format, report file conventions, and the file transfer procedure.

7.2 Trace Specifications

7.2.1 Overview and relationships to IRP Framework

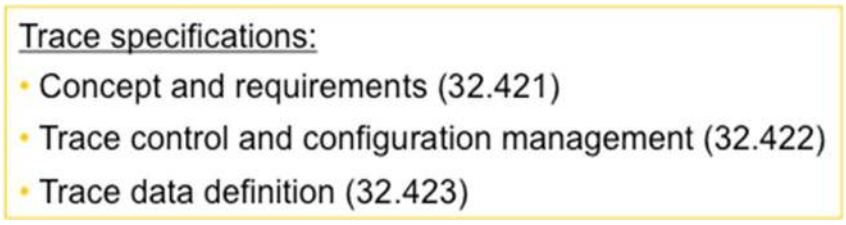


Figure 7.2.1-2: Trace-related Specifications 3GPP R10

7.2.2 Trace Specifications

Trace Control & Configuration – 32.422 [60]

This specification describes the mechanisms used for the control and configuration of the Trace functionality at the EMs, NEs and UEs. It covers the triggering events for starting/stopping of subscriber/UE activity traced over 3GPP standardized signalling interfaces, the types of trace mechanisms, configuration of a trace, level of detail available in the trace data, the generation of Trace results in the Network Elements (NEs) and User Equipment (UE) and the transfer of these results to one or more EM(s) and/or Network Manager(s) (NM(s)).

Trace Data Definitions – 32.423 [61]

This specification describes Trace data definition and management. It covers the trace records content, their format and transfer.

8 Relationships between IRPs

Relationships between Interface IRP's:

- All Interface IRP's utilizing the functionality of the Generic IRP
- All Notification emitting Interface IRP's utilizing the functionality of the Notification IRP

Relationships between NRM IRP's:

- All NRM IRP's utilizing the functionality of the Generic NRM IRP
- All function-specific NRM IRP (e.g. Core NRM IRP) utilizing the functionality of the Inventory NRM IRP

9 SON Functions

In order to reduce the operating expenses (OPEX) associated with the management of increasing number of nodes from more than one vendor the concept of the Self-Organizing Network (SON) was introduced. Automation of some network planning, configuration and optimisation processes via the use of SON functions can help the network operator to reduce OPEX by reducing manual involvement in such tasks.

TS 32.500 [75] identifies the following SON use cases in context of LTE network for standardization:

- Self-establishment of an eNodeB: Related requirements are defined in TS 32.501 [76];
- Automatic Neighbour Relation Management: Related requirements are defined in TS 32.511 [79];
- Self-Optimisation Management: Related requirements are defined in TS 32.521 [80];
- Self-Healing Management: Related requirements are defined in TS 32.541 [86];
- Energy Saving Management: Related requirements are defined in TS 32.551 [87];
- Minimization of Drive Tests: Related requirements are defined in TS 32.421 [59]and TS 32.441 [67].

Each SON use case can involve multiple IRPs when corresponding management scenarios are applied. The IRP relationships of SON functions are addressed case by case as follows,

Self-establishment of an eNodeB

- Generic IRP (32.311[25], 32.312 [26], 32.316 [27])
- Alarm IRP (32.111-1 [5]**,** 32.111-2 [6], 32.111-6 [7])
- Test Management IRP (32.321[28], 32.322[29], 32.326 [30])
- Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
- Kernel CM IRP (32.661 [107], 32.662 [108])
- SC IRP (32.500 [75], 32.501 [76], 32.502 [77], 32.506 [78])
- Software Management IRP (32.531 [83], 32.532 [84], 32.536 [85])
- Notification IRP (32.300 [21], 32.301[22], 32.302 [23], 32.306 [24])
- File Transfer IRP (for ARCF data download) (32.341[34], 32.342[35], 32.346[36])
- Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97])

EUTRAN NRM IRP (32.761 [131], 32.762 [132], 32.766 [133])

Automatic Neighbour Relation Management

- Alarm IRP (32.111-1 [5]**,** 32.111-2 [6], 32.111-6 [5])
- Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
- Kernel CM IRP (32.661 [107], 32.662 [108])
- Notification IRP (32.300 [21], 32.301 [22], 32.302 [23], 32.306 [24])
- Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97] EUTRAN NRM IRP (32.761 [131], 32.762 [132], 32.766 [133])

Self-Optimisation Management

- Alarm IRP (32.111-1 [5], 32.111-2 [6], 32.111-6 [7])

- PM IRP (32.410 [55], 32.411 [56], 32.412 [57], 32.416 [58])
- File Transfer IRP (for PM data upload) (32.341[34], 32.342[35], 32.346[36] Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
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- Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97])
- EUTRAN NRM IRP (32.761 [131], 32.762 [132], 32.766 [133])
- EUTRAN Performance Measurements (32.425 [62])

Self-Healing Management

- Alarm IRP (32.111-1 [5], 32.111-2 [6], 32.111-6 [7])
- Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
- Kernel CM IRP (32.661 [107], 32.662 [108])
- Notification IRP (32.300 [21], 32.301 [22], 32.302 [23], 32.306 [24])
- SON Policy NRM IRP (32.521[80], 32.522 [81], 32.526 [82] Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97])
- EUTRAN NRM IRP (32.761 [131], 32.762 [132], 32.766 [133])

Energy Saving Management

- PM IRP (32.410 [55], 32.411 [56], 32.412 [57], 32.416 [58])
- File Transfer IRP (for PM data upload) (32.341[34], 32.342[35], 32.346[36])
- Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
- Kernel CM IRP (32.661 [107], 32.662 [108])
- Notification IRP (32.300 [21], 32.301 [22], 32.302 [23], 32.306 [24])
- SON Policy NRM IRP (32.521 [80], 32.522 [81], 32.526 [82])
- Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97])
- EUTRAN NRM IRP (32.761 [131], 32.762 [132], 32.766 [133]EUTRAN Performance Measurements (32.425 [62])

Minimization of Drive Tests

- Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
- Kernel CM IRP (32.661 [107], 32.662 [108])
- Trace Management IRP (32.441 [67], 32.442 [68], 32.446 [69])
- Trace Control and Configuration Management IRP (32.421 [59],32.422 [60], 32.423 [61])
- Generic NRM IRP (32.621[95], 32.622 [96], 32.626 [97])
- UTRAN NRM IRP (32.641[101], 32.642 [102], 32.646 [103])
- EUTRAN NRM IRP (32.761 [131], 32.762 [132], 32.766 [133])

Annex A (Informative): IRP Packaging Suggestions

A.1 Overview & Purpose

This Annex provides IRP packaging suggestions, intended to guide service providers as well as equipment and solutions providers to identify and choose suitable standardized management interface capabilities.

Note: No IRP Packaging Suggestions are identified for the present document.

Annex B (informative): Change history

Change history							
Date	**TSG #**	**TSG Doc.**	**CR**	**Rev**	**Subject/Comment**	**Old**	**New**
2010-12	SP-50	SP-100764	--	--	Submitted to SA#50 for Information	0.1.0	1.0.0
2011-05	SP-52	SP-110272	--	--	Submitted to SA#52 for Approval	1.2.0	2.0.0
2011-06	SP-52	--	--	--	Publication	2.0.0	10.0.0

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2020-09	SA#89E	SP-200728	0019	-	F	Correction of reference list and adding references throughout the document	10.1.0

History

Document history		
V10.0.0	July 2011	Publication
V10.1.0	November 2020	Publication